



[ZIMSEC ROBOTICS NOTES]

[FORM 1 TO 6]

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CHAPTER ONE

HISTORY OF ROBOTS

Ancient Times

- The idea of robots dates back thousands of years!
- Around 300 BC, Greek engineer Ctesibius built water-powered machines.
- Leonardo da Vinci (1495) designed a mechanical knight that could sit, wave, and move its jaw.

INDUSTRIAL AGE (1700S–1800S)

- Inventors created automatic machines like music-playing dolls and mechanical clocks.
- These were early "robot-like" machines called automatons .

20TH CENTURY: BIRTH OF MODERN ROBOTS

- 1921 : The word "robot" was first used in a play called R.U.R. by Karel Čapek .
- 1954 : George Devol invented the first programmable robot called Unimate .
- 1961 : Unimate became the first robot used in a factory
- it worked in a General Motors car plant.

21ST CENTURY: SMART AND HELPFUL ROBOTS

- Robots now help in medicine, homes, factories, and space .
- AI (Artificial Intelligence) allows robots to talk, learn, and see.
- Examples: Sophia (humanoid robot), Roomba (robot vacuum), Mars Rovers (space robots)

TODAY AND THE FUTURE

- Robots are being developed for:
 - Teaching in schools
 - Helping the elderly

- Flying (drones)
- Exploring deep oceans or planets

 SUMMARY:

Robots have gone from simple mechanical toys to smart machines that can help, explore, and learn . The journey of robots reflects human creativity and our drive to improve life with technology.

CUBE DOC

CHAPTER TWO

TYPES OF ROBOTS

Here's a simple and clear overview of the types of robots , perfect for students or anyone new to robotics:



TYPES OF ROBOTS

Robots come in many forms depending on their function , design , and environment . Below are the most common types:

1. MOBILE ROBOTS

- Move around to perform tasks
- Use wheels , tracks , or legs
- Common in factories, warehouses, and rescue missions

Examples: Delivery robots, Mars rovers

2. INDUSTRIAL ROBOTS

- Used in factories for repetitive jobs like welding, painting, or assembly
- Usually fixed to one place

Examples : Robotic arms in car factories

3. Service Robots

- Help humans in everyday life
- Used in homes, hospitals, offices

Examples : Robot vacuums (Roomba), hospital delivery robots, humanoid assistants

4. HUMANOID ROBOTS

- Look and sometimes act like humans
- Can walk, talk, and make expressions

Examples : Sophia the robot, ASIMO by Honda

5. MEDICAL ROBOTS

- Assist doctors in surgery, diagnosis, or patient care

- Very precise and reliable

Examples : Da Vinci surgical robot, robotic prosthetic limbs

6. EXPLORATION ROBOTS

- Designed for harsh environments where humans can't go
- Used in space , deep oceans , volcanoes , etc.

Examples : NASA's Mars Rovers, underwater drones

7. MILITARY AND DEFENSE ROBOTS

- Used for surveillance, bomb disposal, or combat support

Examples : Drones, bomb-disposal robots

8. EDUCATIONAL ROBOTS

- Used to teach students programming and robotics
- Often simple, fun, and safe

Examples : LEGO Mindstorms, mBot, Arduino-based kits

9. AGRICULTURAL ROBOTS

- Help farmers with planting, watering, and harvesting

Examples : Crop-spraying drones, autonomous tractors

CHAPTER THREE

BASIC COMPONENTS OF A ROBOT



Basic Components of a Robot

1. STRUCTURE (THE BODY)

- This is the physical frame of the robot.
- Made from materials like plastic, aluminum, or wood.
- Holds all other components together.

Example : Chassis, arms, wheels

2. CONTROLLER (THE BRAIN)

- Controls the robot's actions by processing inputs and sending commands.
- Usually a microcontroller (like Arduino) or a small computer (like Raspberry Pi).

Example : Arduino Uno, Raspberry Pi, ESP32

3. SENSORS (THE EYES AND EARS)

- Help the robot gather information from its surroundings.
- Detects things like distance, light, temperature, or sound.

Common sensors :

- Ultrasonic sensor (for distance)
- IR sensor (for detecting lines or objects)
- Light sensor
- Temperature sensor

4. ACTUATORS (THE MUSCLES)

- Convert electrical signals into movement .
- Allow the robot to move, grab, rotate, or lift.

Types :

- Motors (for wheels)

- Servo motors (for arms or joints)
- Linear actuators (for push/pull movements)

5. POWER SOURCE (THE ENERGY SUPPLY)

- Powers all electronic components.
- Can be batteries or plugged-in power.

Examples :

- AA batteries
- 9V battery
- Rechargeable Li-ion battery pack

6. WIRING AND CONNECTORS

- Connect all components electrically.
- Includes jumper wires, breadboards, or soldered connections.

7. SOFTWARE (THE MIND)

- The code or program that tells the robot what to do .
- Written in languages like C++ , Python , or Scratch .
- Uploaded to the controller to run the robot.

CHAPTER FOUR

SAFETY AND ETHICS OF ROBOTICS

Understanding the safety and ethics of robotics is just as important as building and programming robots.

SAFETY ETHICS IN ROBOTICS

1. Safety in Robotics

Robots can be powerful and sometimes dangerous if not used properly. So, safety is essential for both humans and machines.

Key Safety Considerations:

- **Electrical Safety:** Avoid short circuits, use proper insulation, and don't overload circuits.
- **Mechanical Safety:** Secure moving parts to prevent injury (e.g., spinning wheels, robotic arms).
- **Software Safety :** Write reliable code to avoid unpredictable behavior.
- **Emergency Stop :** Always have a way to shut down the robot in case of a malfunction.
- **Environment Awareness :** Keep robots away from water, heat, or areas with people when testing.

2. ETHICS IN ROBOTICS

Ethics means doing what is right and fair , especially as robots become more intelligent and more involved in daily life.

KEY ETHICAL ISSUES:

- **Privacy :** Robots with cameras or microphones must not invade people's personal space or collect private data without permission.
- **Job Replacement :** As robots take over some jobs, we must think about how to support workers.
- **Bias and Fairness :** Robots powered by AI must not show unfair bias (e.g., in healthcare or education).

- **Responsibility** : Who is to blame if a robot causes harm? The maker, the user, or the programmer?

3 LAWS OF ROBOTICS (ASIMOV'S RULES)

Fictional but widely discussed ethical rules:

1. A robot may not harm a human being.
2. A robot must obey human orders, unless it conflicts with rule 1.
3. A robot must protect itself, unless it conflicts with rules 1 or 2.

SUMMARY

Robots must be:

- Safe to use
- Ethically designed to respect human rights
- Used responsibly in homes, schools, hospitals, and workplaces

CHAPTER FIVE

SENSORS AND ACTUATORS

SENSORS

What Are Sensors in Robotics?

Sensors are the "eyes, ears, and skin" of a robot. They help robots detect and understand their surroundings by collecting real-world data such as distance, light, sound, heat, or touch.

Common Types of Sensors in Robotics

Sensor Type	What It Detects	Common Use in Robots
Ultrasonic Sensor	Distance to an object	Obstacle detection, parking sensors
Infrared (IR) Sensor	Proximity, line detection	Line-following robots, object detection
Light Sensor	Light intensity	Light-seeking or light-avoiding robots
Touch Sensor	Physical contact (pressure)	Bump detection or collision sensing
Temperature Sensor	Heat or cold	Environmental monitoring
Sound Sensor	Noise or sound waves	Voice-activated robots
Gyroscope	Orientation/rotation	Balancing robots, drones
GPS Sensor	Location on Earth	Navigation for outdoor robots or drones
Accelerometer	Movement and speed changes	Motion detection, tilt sensors
Camera (Vision Sensor)	Images or video	Object recognition, facial detection

EXAMPLE:

How an Ultrasonic Sensor Works

- Sends out a sound wave (like a bat).
- Waits for it to bounce back from an object.
- Measures how long it took → calculates distance.

WHY ARE SENSORS IMPORTANT?

- They help the robot understand its environment.
- Allow robots to make decisions (e.g., stop if something is in the way).
- Enable autonomous behavior.

ACTUATORS

WHAT ARE ACTUATORS IN ROBOTICS?

Actuators are the "muscles" of a robot. They are components that convert electrical energy into physical motion, allowing the robot to move, rotate, lift, or grab things.

MAIN TYPES OF ACTUATORS

Type of Actuator	Movement Produced	Common Uses in Robots
DC Motor	Continuous rotation	Wheels, fans, or spinning parts
Servo Motor	Precise angle rotation	Robotic arms, grippers, joints
Stepper Motor	Step-by-step rotation	3D printers, CNC machines, precise control
Linear Actuator	Push or pull (straight line)	Lifting or sliding actions
Pneumatic Actuator	Movement using air pressure	Fast motion in factory robots
Hydraulic Actuator	Movement using liquid pressure	Heavy-duty robots and machines

EXAMPLE:

A servo motor can rotate an arm exactly 90 degrees to pick up an object, while a DC motor can make a robot's wheels spin so it moves forward.

WHY ARE ACTUATORS IMPORTANT?

- They bring the robot to life by creating movement .
- Without actuators, the robot would just be a static machine.
- They are controlled by code to react to sensor input.

SIMPLE PROJECT EXAMPLE:

- Wheels + DC Motors + Arduino = Mobile Robot
- Servo + Ultrasonic Sensor = Scanning head to detect objects

CHAPTER SIX

TOOLS AND APPLICATIONS OF ROBOT DESIGN

TOOLS OF ROBOT DESIGN

TOOL		
Arduino IDE	Programming microcontrollers	Writing code for motors and sensors
Tinkercad Circuits	Virtual circuit simulation (online)	Testing circuits before building
Fritzing	Creating wiring diagrams	Visualizing sensor/motor connections
Scratch / mBlock	Block coding for beginners	Programming simple robots visually
SolidWorks / Fusion 360	CAD (3D Design)	Designing robot body or frame
ROS (Robot Operating System)	Advanced control system framework	For autonomous or AI-based robots

3. PROGRAMMING LANGUAGES FOR ROBOTS

1. C/C++ - Arduino-based robots |
2. Python - AI, Raspberry Pi, vision systems |
3. Blockly/Scratch -Beginners & educational robots |
4. MATLAB -Advanced simulations and control |

APPLICATIONS FOR DESIGNING ROBOTS

These tools help you design , simulate , and program robots effectively.

1. CAD SOFTWARE (MECHANICAL DESIGN)

Used to design the robot's physical structure (chassis, arms, joints).

Application	Purpose	Notes
Fusion 360	3D design, simulation	Free for students; cloud-based
SolidWorks	Detailed mechanical CAD	Widely used in industry
Tinkercad	Simple 3D and circuit design	Great for beginners
AutoCAD	2D/3D mechanical design	Professional-grade

2. SIMULATION SOFTWARE (VIRTUAL TESTING)

Simulate robot movement, sensors, and environments without building a real robot.

Application	Purpose	Notes
Gazebo	Physics-based robot simulation	Works with ROS
Webots	Open-source 3D robot simulation	Good for education and research
V-REP (CoppeliaSim)	Robot modeling and simulation	Advanced simulations
Tinkercad Circuits	Simulate Arduino + circuits	Great for beginners

3. PROGRAMMING ENVIRONMENTS

Used to code your robot's brain and behavior.

Application	Purpose	Notes
Arduino IDE	Programming microcontrollers	Hardware-based robots
ROS (Robot Operating System)	Middleware for robot control	Advanced robotics projects

Python	High-level coding, AI, sensors	Raspberry Pi, vision robots
Scratch / mBlock	Block-based programming	Young learners and beginners

4. WIRING & CIRCUIT DESIGN TOOLS

Helps you plan and document the electronics inside your robot.

Application	Purpose
Fritzing	Create wiring diagrams
Tinkercad Circuits	Simulate electronics + Arduino
KiCad	PCB design for advanced builders

5. GAME ENGINES (FOR ROBOTICS + AI SIMULATION)

Used for simulating robot environments or integrating AI.

Application	Purpose
Unity 3D	Game engine for robot AI and visuals
Unreal Engine	Realistic robot simulation environments

CHAPTER SEVEN

ROBOT CONSTRUCTION AND PROGRAMMING

Constructing a robot can be a fun and rewarding project! Here's a step-by-step guide to building a simple mobile robot (e.g., an obstacle-avoiding robot), suitable for beginners using tools like Arduino.

HOW TO CONSTRUCT A SIMPLE ROBOT

Step 1: Plan Your Robot

Decide what your robot will do. For a beginner, a robot that:

- Moves forward
- Avoids obstacles
- Uses sensors and wheels

Step 2: Gather Materials

Essential components:

- Microcontroller : Arduino Uno or similar
- Chassis : Frame or base (can be 3D-printed or a kit)
- 2 DC motors + wheels
- Motor driver (e.g., L298N)
- Ultrasonic sensor (for obstacle detection)
- Battery pack (e.g., 9V or Li-ion)
- Jumper wires , breadboard, screws

Optional:

- Caster wheel (for balance)
- LED, buzzer (for signals)

Step 3: Assemble the Chassis

- Mount the motors to the chassis using brackets or glue.

- Attach the wheels to the motor shafts.
- Fix the caster wheel at the front or back for stability.

Step 4: Wire the Electronics

Connect everything according to your robot's design:

- Motors → Motor Driver
- Motor Driver → Arduino
- Ultrasonic Sensor → Arduino (Trig/Echo pins)
- Battery → Arduino and Motor Driver

NB: Use a breadboard for testing before soldering.

Step 5: Program the Robot

Upload this sample Arduino code (obstacle avoidance):

```
```cpp
#define trigPin 9
#define echoPin 10
#define motor1A 2
#define motor1B 3
#define motor2A 4
#define motor2B 5

void setup() {
 pinMode(trigPin, OUTPUT);
 pinMode(echoPin, INPUT);
 pinMode(motor1A, OUTPUT);
 pinMode(motor1B, OUTPUT);
 pinMode(motor2A, OUTPUT);
```

```
pinMode(motor2B, OUTPUT);
}
void loop() {
long duration, distance;
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
duration = pulseIn(echoPin, HIGH);
distance = duration * 0.034 / 2;

if (distance < 15) {
 // Obstacle detected → turn
 digitalWrite(motor1A, LOW);
 digitalWrite(motor1B, HIGH);
 digitalWrite(motor2A, HIGH);
 digitalWrite(motor2B, LOW);
 delay(500);
} else {
 // Move forward
 digitalWrite(motor1A, HIGH);
 digitalWrite(motor1B, LOW);
 digitalWrite(motor2A, HIGH);
 digitalWrite(motor2B, LOW);
}
```

```
}

}
```

### **Step 6: Power and Test**

- Connect your battery pack.
- Place the robot on the ground.
- Observe how it moves and responds to obstacles.

### **Step 7: Improve and Upgrade**

- Add more sensors (e.g., IR, light).
- Add a Bluetooth or Wi-Fi module.
- Build a remote control or use a phone

## **ROBOT PROGRAMMING**

Robot programming is the process of writing instructions that tell a robot how to think , move , and respond to its environment. It's how we make robots come to life!

### **What Is Robot Programming?**

It involves:

- Controlling hardware like motors and sensors
- Processing sensor data (e.g., distance, light, sound)
- Making decisions (e.g., if an obstacle is near, turn away)
- Following commands or running autonomously

### **Common Programming Languages for Robots**

### **Example: Arduino Robot Program**

This code makes a robot move forward and stop if something is close:

```
```cpp
```

```
int trigPin = 9;
```

```
int echoPin = 10;
```

```
int motor1 = 2;
```

```
int motor2 = 3;
```

```
void setup() {
```

```
    pinMode(trigPin, OUTPUT);
```

```
[05/05, 13:05] ChatGPT: pinMode(echoPin, INPUT);
```

```
    pinMode(motor1, OUTPUT);
```

```
    pinMode(motor2, OUTPUT);
```

```
}
```

```
void loop() {
```

```
    digitalWrite(trigPin, LOW);
```

```
    delayMicroseconds(2);
```

```
    digitalWrite(trigPin, HIGH);
```

```
    delayMicroseconds(10);
```

```
    digitalWrite(trigPin, LOW);
```

```
    int duration = pulseIn(echoPin, HIGH);
```

```
    int distance = duration * 0.034 / 2;
```

```
    if (distance < 15) {
```

```

digitalWrite(motor1, LOW);

digitalWrite(motor2, LOW);

} else {

    digitalWrite(motor1, HIGH);

    digitalWrite(motor2, HIGH);

}

}

```

STEPS IN ROBOT PROGRAMMING

1. Write the code on a computer (using Arduino IDE, Python, or Scratch)
2. Upload it to the robot's controller (e.g., Arduino)
3. Test the behavior (movement, sensing, decision-making)
4. Debug and improve until the robot performs the desired tasks

Programming Tools

TOOL	Purpose
Arduino IDE	Code for Arduino robots
mBlock/Scratch	Visual programming for kids
Thonny/Python	Code for Raspberry Pi robots
Webots, Gazebo	Simulate robots before building

Examples

Great! Let's start with a beginner-friendly robot programming project using Arduino — ideal for small robots or school projects.

Beginner Arduino Robot Project: LED Blinking (Robot's First Signal)

This project teaches you the basics of coding and controlling a robot's output (like an LED, which can act as a signal light).

What You Need:

- 1 × Arduino Uno
- 1 × LED
- 1 × 220-ohm resistor
- Jumper wires
- Breadboard
- USB cable

Circuit Setup:

1. Connect long leg of LED to Arduino pin 13.
2. Connect short leg of LED through a resistor to GND.
3. Plug in the Arduino to your computer using a USB cable.

Arduino Code:

```
```cpp
void setup() {
 pinMode(13, OUTPUT); // Set pin 13 as an output
}

void loop() {
 digitalWrite(13, HIGH); // Turn LED on
 delay(1000); // Wait 1 second
 digitalWrite(13, LOW); // Turn LED off
 delay(1000); // Wait 1 second
}
```

### **What This Does:**

- Makes the LED blink on and off every second .
- It's the "Hello, World!" of robot programming.

Once you're confident with this, we can move on to:

- Controlling motors
- Reading sensor input
- Making an obstacle-avoiding robot

### Example 2

Let's now build and program a simple Arduino-powered robot that moves forward and turns when it detects an obstacle using an ultrasonic sensor and motors .

### **Obstacle-Avoiding Robot Using Arduino**

#### **What You Need:**

- Arduino Uno
- 2× DC motors (with wheels)
- 1× Motor driver (e.g. L298N)
- 1× Ultrasonic sensor (HC-SR04)
- 1× Caster wheel (for balance)
- Jumper wires
- 1× Battery pack (6V–12V)
- Robot chassis (optional: cardboard or plastic frame)

#### **Wiring Summary:**

Motors → Motor Driver → Arduino:

- Motor A: Out1, Out2 → Left motor



- Motor B: Out3, Out4 → Right motor
- IN1, IN2 → Arduino pins 2, 3
- IN3, IN4 → Arduino pins 4, 5

*Ultrasonic Sensor:*

- VCC → 5V
- GND → GND
- Trig → Pin 9
- Echo → Pin 10

 Arduino Code:

```
```cpp
#define trigPin 9
#define echoPin 10
#define IN1 2
#define IN2 3
#define IN3 4
#define IN4 5

void setup() {
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);
```

```
pinMode(IN3, OUTPUT);  
pinMode(IN4, OUTPUT);  
Serial.begin(9600);  
}
```

```
void loop() {  
    long duration;  
    int distance;
```

```
    digitalWrite(trigPin, LOW);  
    delayMicroseconds(2);  
    digitalWrite(trigPin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trigPin, LOW);  
    duration = pulseIn(echoPin, HIGH);  
    distance = duration * 0.034 / 2;
```

```
    Serial.print("Distance: ");  
    Serial.println(distance);
```

```
    if (distance < 15) {  
        // Turn  
        digitalWrite(IN1, LOW);  
        digitalWrite(IN2, HIGH);  
        digitalWrite(IN3, HIGH);
```

```
digitalWrite(IN4, LOW);  
delay(500);  
} else {  
    // Move forward  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
}  
}
```

How It Works:

- The robot moves forward .
- If an object is detected within 15 cm , it turns to avoid it.